BALA'S A141057 SUPERCONGRUENCE AT INTEGER INDICES

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Abstract. The number of Abelian cubes of length $3n$ over a three-letter alphabet is $a(n) = \sum_{k=0}^{n} \binom{n}{k}^3 \sum_{j=0}^{k} \binom{k}{j}^3$. Bala conjectured in 2022 that its extension to negative indices, obtained by the same finite sum with upper limit $|n|$ and generalized binomial coefficients, satisfies $a(np^r) \equiv a(np^{r-1}) \pmod{p^{3r}}$ for every nonzero integer n, prime $p \geq 5$, and $r \geq 1$. We prove this assertion. More generally, powers $q \geq 3$ of the signed coefficients $\binom{n}{j+\ell}\binom{j+\ell}{j}$ satisfy a coefficientwise Frobenius congruence with modulus $p^{3(v_p(n)+1)}$. The negative-index step follows from an exact normalization by an ordinary multinomial coefficient: its additional rational factor is unchanged by simultaneous dilation. Integral index-multiplied identities supply the required valuation bounds. We include an elementary proof of the multinomial scaling congruence used in the argument.

1. Introduction

An Abelian cube is a concatenation $xx'x''$ in which the three words have equal lengths and the same multiplicity of each letter. The number of such words of length $3n$ over an alphabet of size three is the sequence A141057 in the On-Line Encyclopedia of Integer Sequences [5]. Its two equivalent formulas are

$$a(n) = \sum_{k=0}^{n} \binom{n}{k}^3 \sum_{j=0}^{k} \binom{k}{j}^3 = \sum_{\substack{b_1+b_2+b_3=n \\ b_i \geq 0}} \binom{n}{b_1, b_2, b_3}^3, \qquad n \geq 0. \tag{1}$$

In April 2022, Peter Bala proposed the congruence

$$a(np^r) \equiv a(np^{r-1}) \pmod{p^{3r}} \tag{2}$$

for primes $p \geq 5$ and positive integers n, r. He also observed that the finite negative-index extension appears to satisfy the same congruences. The positive-index statement has a Lean proof reported by Adamczewski [1]; the published proof submission [3] explicitly omits the negative-index extension. In the inspected DeepMind repository [2], the positive statement is marked solved and the integer-indexed statement is designated `OeisA141057.conjecture2`.

For all $N \in \mathbb{Z}$ define

$$a(N) := \sum_{k=0}^{|N|} \binom{N}{k}^3 \sum_{j=0}^{k} \binom{k}{j}^3, \qquad \binom{-m}{k} = (-1)^k \binom{m+k-1}{k} \quad (m \geq 1). \tag{3}$$

The first negative-index values are $-1, 255, -53893, 14396623, -4388536251$. The cutoff $|N|$ is part of the definition; no infinite hypergeometric continuation is involved.

Theorem 1.1 (Bala's integer-indexed supercongruence)**.** *Let $p \geq 5$ be prime, $n \in \mathbb{Z} \setminus \{0\}$, and $r \geq 1$. The sequence defined by* (3) *satisfies* (2)*. In fact, its difference is divisible by $p^{3(v_p(n)+r)}$.*

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The proof retains the familiar division of multinomial indices according to their divisibility by p. The additional observation is that negative coefficients admit a normalization whose rational factor cancels exactly under dilation. This avoids treating that factor as a unit, which would lose the essential valuation information. We prove a coefficientwise statement in Section 4. No claim of publication priority is made for the negative-index extension.

2. An elementary multinomial scaling lemma

Fix a prime $p \geq 5$. We use the valuation v_p on nonzero rational numbers, extend it by $v_p(0) = \infty$, and write $\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}$. Congruences in this ring mean valuation inequalities. For a nonnegative integer vector $\mathbf{b} = (b_1, \ldots, b_d)$, put

$$M(\mathbf{b}) := \frac{(b_1 + \cdots + b_d)!}{b_1! \cdots b_d!}.$$

Lemma 2.1. *Suppose that every b_i is divisible by p^s, where $s \geq 0$. Then*

$$\frac{M(p\mathbf{b})}{M(\mathbf{b})} \equiv 1 \pmod{p^{3(s+1)}}. \tag{4}$$

The quotient is a unit of $\mathbb{Z}_{(p)}$. Zero components, including the all-zero vector, are allowed.

Proof. Put $h = s + 1$ and $L = p^h$. Let I be the integers i with $1 \leq i < L$ and $p \nmid i$, and set $S_1 = \sum_{i \in I} i^{-1}$ and $S_2 = \sum_{i \in I} i^{-2}$. Multiplication by 2 permutes the units modulo L. Consequently $S_2 \equiv \frac{1}{4} S_2 \pmod{L}$; since 3 and 4 are units,

$$S_2 \equiv 0 \pmod{L}. \tag{5}$$

The substitution $i \mapsto L - i$ and the identity

$$\frac{1}{L - i} \equiv -\frac{1}{i} - \frac{L}{i^2} \pmod{L^2}$$

give $2S_1 \equiv -LS_2 \equiv 0 \pmod{L^2}$. Thus $S_1 \equiv 0 \pmod{L^2}$.

Expand the finite product for any integer $t \geq 0$:

$$\begin{aligned} \prod_{i \in I} \left(1 + \frac{tL}{i}\right) &\equiv 1 + tLS_1 + \frac{t^2 L^2}{2}(S_1^2 - S_2) \pmod{L^3} \\ &\equiv 1 \pmod{L^3}. \end{aligned}$$

Every coefficient of degree at least three belongs to $\mathbb{Z}_{(p)}$, so its term is divisible by L^3. It follows that

$$U(t) := \prod_{i \in I} (tL + i) \equiv U(0) \pmod{L^3}.$$

Let $F_p(z) = \prod_{1 \leq i \leq z,\ p \nmid i} i$, with $F_p(0) = 1$. Splitting into complete blocks yields

$$F_p(Lt) \equiv U(0)^t \pmod{L^3}. \tag{6}$$

For $B = \sum_i b_i$, the factorial identity $(pb)! = p^b b! F_p(pb)$ gives the exact equality

$$\frac{M(p\mathbf{b})}{M(\mathbf{b})} = \frac{F_p(pB)}{\prod_i F_p(pb_i)}.$$

Every endpoint is divisible by L. In (6), the numerator exponent is $B/p^s = \sum_i b_i/p^s$, exactly the sum of the denominator exponents. The factors being canceled are units. This proves (4); the all-zero case is the identity $1 = 1$. □

Lemma 2.1 is the common-divisor part of the classical Ljunggren–Jacobsthal congruence; see, for example, [4]. The proof above records precisely the portion needed here.

3. Signed coefficients and their dilation

For $N \neq 0$ and nonnegative j, ℓ with $j + \ell \leq |N|$, write

$$T_N(j,\ell) := \binom{N}{j+\ell}\binom{j+\ell}{j}. \tag{7}$$

For $N > 0$ this is $M(N-j-\ell, j, \ell)$.

Lemma 3.1. *If $(j,\ell) \neq (0,0)$ and $u = \min(v_p(j), v_p(\ell))$, then*

$$v_p(T_N(j,\ell)) \geq \max\{0, v_p(N) - u\}. \tag{8}$$

Proof. For $j > 0$, the two standard binomial identities give

$$jT_N(j,\ell) = N\binom{N-1}{j+\ell-1}\binom{j+\ell-1}{j-1}. \tag{9}$$

For $\ell > 0$ there is the symmetric identity

$$\ell T_N(j,\ell) = N\binom{N-1}{j+\ell-1}\binom{j+\ell-1}{j}.$$

These are identities of integers for both signs of N. Apply the identity whose positive index has valuation u. Integrality supplies the bound zero, and taking valuations supplies the other bound in (8). □

Lemma 3.2. *Let $(j,\ell) \neq (0,0)$ and put $s = \min\{v_p(N), v_p(j), v_p(\ell)\}$. Then*

$$R := \frac{T_{pN}(pj, p\ell)}{T_N(j,\ell)} \equiv 1 \pmod{p^{3(s+1)}}. \tag{10}$$

In particular, R is a unit of $\mathbb{Z}_{(p)}$.

Proof. If $N > 0$, all components of $(N-j-\ell, j, \ell)$ are divisible by p^s. Lemma 2.1 applies directly.

If $N = -m < 0$, with $k = j + \ell$, then

$$\begin{aligned} T_{-m}(j,\ell) &= (-1)^k \frac{(m+k-1)!}{(m-1)!j!\ell!} \\ &= (-1)^k \frac{m}{m+k} M(m,j,\ell). \end{aligned} \tag{11}$$

The fraction $m/(m+k)$ is unchanged under simultaneous dilation of (m,j,ℓ) by p. Since p is odd, the sign is unchanged too. Hence

$$\frac{T_{-pm}(pj,p\ell)}{T_{-m}(j,\ell)} = \frac{M(pm,pj,p\ell)}{M(m,j,\ell)}.$$

All three components are divisible by p^s, and Lemma 2.1 proves (10). The fraction in (11) was canceled as an exact nonzero rational number; no assertion that it is a p-adic unit is needed. □

4. The coefficientwise theorem

For an integer $q \geq 3$, define

$$P_{N,q}(X,Y) := \sum_{\substack{j,\ell \geq 0 \\ j+\ell \leq |N|}} T_N(j,\ell)^q X^j Y^\ell \in \mathbb{Z}[X,Y].$$

At $q = 3$ and $X = Y = 1$, reindexing $k = j + \ell$ gives $P_{N,3}(1,1) = a(N)$.

Theorem 4.1. *For every integer $N \neq 0$, prime $p \geq 5$, and integer $q \geq 3$,*

$$P_{pN,q}(X,Y) \equiv P_{N,q}(X^p, Y^p) \pmod{p^{3(v_p(N)+1)}} \tag{12}$$

coefficientwise in $\mathbb{Z}[X,Y]$.

Proof. Write $e = v_p(N)$. Consider first a pair (J, K) for which at least one index is not divisible by p. Lemma 3.1, with pN in place of N, gives $v_p(T_{pN}(J, K)) \geq e+1$. Its q-th power vanishes modulo $p^{3(e+1)}$.

The remaining pairs are $(J, K) = (pj, p\ell)$, and $J + K \leq |pN|$ is equivalent to $j + \ell \leq |N|$. For $(j, \ell) = (0, 0)$, both coefficients are 1. Otherwise let $s = \min\{e, v_p(j), v_p(\ell)\}$ and use R from Lemma 3.2. Since $s \leq e$, Lemma 3.1 gives $v_p(T_N(j, \ell)) \geq e - s$. The factorization $R^q - 1 = (R-1)\sum_{i=0}^{q-1} R^i$ then gives

$$\begin{aligned} v_p\big(T_{pN}(pj, p\ell)^q - T_N(j, \ell)^q\big) &\geq q(e-s) + 3(s+1) \\ &= 3(e+1) + (q-3)(e-s) \\ &\geq 3(e+1). \end{aligned}$$

If the difference is zero the same inequality holds with valuation ∞. Every coefficient has been treated, proving (12). □

Proof of Theorem 1.1. Apply Theorem 4.1 at $q = 3$, $X = Y = 1$, and $N = np^{r-1}$. Its modulus is $p^{3(v_p(n)+r)}$, which is divisible by p^{3r}. This proves (2) with the stated strengthening. □

Remark 4.2. The endpoint of the negative sum causes no additional terms: the equivalence $pj+p\ell \leq p|N| \iff j + \ell \leq |N|$ includes equality. Neither n nor the positive indices need be prime to p. The convention $v_p(0) = \infty$ accommodates one zero index, while $(0, 0)$ is handled separately. No conclusion about $p = 2$ or $p = 3$ is asserted here.

5. Formal statement and reproducibility

The formal proof is contained in the single source file `BalaA141057Conjecture.lean`. It uses Lean 4.33.1 and the corresponding Mathlib environment. The target is the exact declaration `OeisA141057.conjecture2` of [2], with the source definition `aInt`; in particular, its negative sum uses the cutoff $(-n)$.`toNat` and the sign $((-1)^k\binom{-n+k-1}{k})^3$. The positive-case helpers from [3] are included in the file with attribution, rather than importing the repository's unproved conjecture as a premise.

The theorem `BalaA141057.integer_supercongruence` has passed Lean's kernel check. A verification command compares its type directly with the repository declaration, including all hypotheses and the source definition of the sequence. A second command rejects any axiom other than `propext`, `Classical.choice`, and `Quot.sound`; these are precisely the reported dependencies. The proof contains no admissions and does not depend on the repository's admitted conjectures. The accompanying verification report records the source hashes, compiler version, and pinned library revisions. The formal result is the exact scalar congruence (2); the general polynomial statement of Theorem 4.1 is proved in the mathematical argument above.

An independent exact-integer checker evaluates the source's nested finite sums. It verifies 168 full congruences for both signs, including $p \mid n$, levels through $r = 4$, and absolute indices through 625. It also verifies 180,306 exhaustive coefficient comparisons, 10,430 individual coefficient comparisons, and 23 unit-block regimes. These finite computations check the implementation of the formulas; the unrestricted theorem follows from the preceding proof.

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